

A RANK METHOD FOR COMPLEMENTS OF FINITE COSET UNIONS

ABSTRACT. We prove an affirmative answer to Kourovka Notebook Problem 21.115: if $C_1, \dots, C_n$ are arbitrary independently left or right cosets, possibly of different nonnormal subgroups of a finite group G , and $G \setminus \bigcup_i C_i$ is nonempty, then it has size at least $|G|/2^n$. The proof constructs rank-two matrices whose Hadamard product has exactly the complement Cayley support and compares its rank with a sparse-transversal lower bound. The same matrix yields left and right translate covers with at most 2^n translates and shows that equality forces the uncovered set to be a subgroup coset of index 2^n . For compatible blocks of b_j left or right cosets of a common subgroup, a univariate detector of rank at most $b_j + 1$ gives the sharper denominator $\prod_j (b_j + 1)$, with corresponding translate-cover and equality statements. We record sharp examples and several compact consequences.

1. INTRODUCTION AND STATEMENT OF THE PROBLEM

Problem 21.115 first appeared in the official version 39 of the 21st *Kourovka Notebook*, dated 8 January 2026, on printed page 178 [KM26a, p. 178]. It is attributed there to B. Sambale and asks:

Let $C_1, \dots, C_n$ be (left or right) cosets of a finite group G such that $U = C_1 \cup \dots \cup C_n$ is not G . Is it always true that $|G \setminus U| \geq |G|/2^n$?

The wording permits the orientation to be chosen independently for each coset; the underlying subgroups may differ and need not be normal. In the current checked version 45, dated 3 July 2026, the problem appears on printed page 181 with unchanged wording and no solution annotation [KM26b, p. 181]. The same bound had been formulated conjecturally by Sambale and Tărnăuceanu, who proved a general $|G|/(2n!)$ lower bound and the desired estimate in several special cases [ST22].

Our main result is the exact affirmative answer.

Theorem 1.1 (Mixed-coset complement bound). *Let $n \geq 1$, let G be a finite group, and let $C_1, \dots, C_n$ be sets such that each C_i is a left or right coset of a subgroup of G . The orientations, subgroups, and repetitions may be arbitrary. If*

$$A := G \setminus \bigcup_{i=1}^n C_i \neq \emptyset,$$

then

$$|A| \geq \frac{|G|}{2^n}.$$

Equivalently, $|A| \geq \lceil |G|/2^n \rceil$.

The proof is an exact rank sandwich. One constructs a matrix M with $M(x, y) \neq 0$ precisely when $x^{-1}y \in A$, proves $\text{rank } M \leq 2^n$, and uses the support of an actual row basis to prove $\text{rank } M \geq |G|/|A|$. This also produces a nonsingular submatrix whose nonzero entries are all supported on the complement relation.

The same argument gives two structural strengthenings. First, at most 2^n left translates of A cover G , and the same is true on the right. Second, equality in Theorem 1.1 forces A to be a subgroup coset of index 2^n . When several forbidden cosets share one subgroup and one orientation, their individual rank-two factors can be compressed into a single rank- $(b+1)$ block. A targeted search through 2 August 2026 did not locate this exact mixed-orientation

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compatible-block construction or its product-denominator consequences; it is therefore presented only as an apparently new application, with moderate confidence and no claim of historical priority.

Several ingredients and later consequences have established antecedents. Hadamard-rank submultiplicativity is standard [Joh78, Observation 7, p. 587] and [AJS09, Proposition 2, p. 27]; the sparse support lemmas below are convenient elementary formulations of basis and minor facts, with close support–dimension analogues in permutation modules [GGI05]. The binary subgroup-chain example is a special case of Sun’s exact-cover construction [Sun01, Example 1.2]; see also [Sun06]. The exact-support parameter used in the appendix is the Cayley specialization of established zero-pattern and nondeterministic-rank notions [HS93; Wol00]. The finite-index and profinite statements are routine quotient transfers. None of these observations is a priority certificate for the main theorem or the compatible-block refinement.

Two distinct audited Lean 4/Mathlib certificates verify the headline mixed-left/right finite-group complement inequality: an independently written Lean 4.30 proof and an Aristotle reconstruction with an audited one-line compatibility port to Lean 4.30. Neither certificate formalizes sharpness, translate-cover conclusions, equality rigidity, compatible grouped blocks, exact-support rank consequences, prescribed-image applications, or the finite-index and profinite extensions.

The 2022 special cases. The following facts were already established in [ST22]. Proposition 2 gives the general estimate $|G \setminus \bigcup_i C_i| \geq |G|/(2n!)$. Proposition 3 proves the target 2^{-n} estimate for unions of subgroups when $n \leq 5$, and Lemma 4 reduces the nilpotent case to finite p -groups. Theorem 6 gives an optimal formula for elementary abelian p -groups: if $n = a + b(p - 1)$ with $0 \leq a \leq p - 2$, then the uncovered proportion is at least $(p - a)/p^{b+1}$. The paper also obtains the target bound for finite p -groups when $n \leq 2p - 2$ and for cyclic groups. The argument below is different and treats arbitrary finite groups and arbitrary mixed left/right subgroup cosets.

2. TWO ELEMENTARY RANK LEMMAS

All ranks are ordinary matrix ranks over the stated field. If matrices have the same shape, $B \circ D$ denotes their entrywise, or Hadamard, product.

Lemma 2.1 (Hadamard-rank inequality). *Let $s \geq 1$, and let $B_1, \dots, B_s$ be matrices of the same finite shape over a field $\mathbb{F}$. Then*

$$\text{rank}(B_1 \circ \dots \circ B_s) \leq \prod_{i=1}^s \text{rank } B_i.$$

Proof. Write $B_i = \sum_{j=1}^{r_i} u_{ij} v_{ij}^\top$, where $r_i = \text{rank } B_i$. Distributing the entrywise product expresses $B_1 \circ \dots \circ B_s$ as a sum of $\prod_i r_i$ outer products, since

$$(u_1 v_1^\top) \circ \dots \circ (u_s v_s^\top) = (u_1 \circ \dots \circ u_s)(v_1 \circ \dots \circ v_s)^\top.$$

Each summand has rank at most one. This is the standard Hadamard-rank bound; see also [Joh78, Observation 7, p. 587] and [AJS09, Proposition 2, p. 27]. $\square$

Lemma 2.2 (Sparse transversal and supported minor). *Let B be an $N \times N$ matrix over a field, where $N > 0$. Suppose every row has at most m nonzero entries and there is a permutation π of the column set such that $B(x, \pi(x)) \neq 0$ for every row x . Then, writing $r = \text{rank } B$,*

$$N \leq mr.$$

Moreover, r actual row supports cover all columns, r actual column supports cover all rows, and there are actual row and column index sets X, Y with $|X| = |Y| = r$ such that $B[X, Y]$ is nonsingular.

Proof. Choose r actual rows forming a basis of the row space. If a column vanished on those rows, it would vanish on every row. This is impossible because the transversal supplies a nonzero entry in every column. Thus the basis-row supports cover all N columns, and $N \leq mr$. Applying the same argument to columns gives the transposed support-cover assertion. Finally, the $r \times N$

matrix formed by the basis rows has row rank r , so it has r pivot columns; the corresponding $r \times r$ submatrix is nonsingular. $\square$

In particular, a square matrix with nonzero diagonal and at most m nonzeros per row has rank at least N/m . These statements are elementary basis and rank-minor facts; they are recorded in this form only because every part is used below.

3. THE RANK-TWO DETECTOR AND THE PROOF

3.1. Orientation and normalization. For every subgroup $H \leq G$ and $a \in G$,

$$Ha = a(a^{-1}Ha).$$

Indeed, both sets consist of the elements ha with $h \in H$. Hence every right coset is a left coset of a conjugate subgroup, and this conversion may be made independently for each C_i . We may write $C_i = a_i H_i$ without assuming that any H_i is normal.

Choose $g_0 \in A$ and left-translate the entire configuration by g_0^{-1} . Each $a_i H_i$ remains a left coset, cardinalities are unchanged, and the translated complement contains the identity. After renaming, we assume

$$e \in A, \quad C_i = a_i H_i, \quad a_i \notin H_i.$$

The last condition is equivalent to $e \notin a_i H_i$.

3.2. One forbidden coset. Regard G/H_i only as the finite set of left cosets and choose an injection $\lambda_i : G/H_i \rightarrow \mathbb{Q}$. Define a $G \times G$ matrix by

$$M_i(x, y) = \lambda_i(yH_i) - \lambda_i(xa_i H_i).$$

Writing $\alpha_i(y) = \lambda_i(yH_i)$ and $\beta_i(x) = \lambda_i(xa_i H_i)$ shows that $M_i(x, y) = \alpha_i(y) - \beta_i(x)$, the difference of two outer products; therefore $\text{rank } M_i \leq 2$. Injectivity gives the exact zero pattern

$$\begin{aligned} M_i(x, y) = 0 &\iff yH_i = xa_i H_i \\ &\iff x^{-1}y \in a_i H_i = C_i. \end{aligned}$$

No quotient-group operation is used. In particular, $M_i(x, x) \neq 0$ because $e \notin C_i$.

3.3. The complement matrix. Set

$$M = M_1 \circ \cdots \circ M_n.$$

Since $\mathbb{Q}$ has no zero divisors,

$$(3.1) \quad M(x, y) \neq 0 \iff x^{-1}y \notin C_i \text{ for every } i \iff x^{-1}y \in A.$$

Thus row x has support xA and exactly $|A|$ nonzero entries, while every diagonal entry is nonzero. By Lemma 2.1, $\text{rank } M \leq 2^n$. Equivalently, direct expansion gives

$$(3.2) \quad M(x, y) = \sum_{S \subseteq [n]} (-1)^{|S|} \left(\prod_{i \in S} \beta_i(x) \right) \left(\prod_{i \notin S} \alpha_i(y) \right),$$

a sum of 2^n outer products.

Apply Lemma 2.2 with the diagonal transversal. With $N = |G|$ and $m = |A|$,

$$\frac{|G|}{|A|} \leq \text{rank } M \leq 2^n,$$

which proves Theorem 1.1 in the normalized configuration. Undoing the left translation preserves $|A|$ and completes the proof.

The same lemma supplies an explicit internal certificate: there are $X, Y \subseteq G$ with $|X| = |Y| = \text{rank } M$ such that $M[X, Y]$ is nonsingular. Every nonzero entry of this minor satisfies $x^{-1}y \in A$ by Equation (3.1). The minor is not asserted to be principal or to equal $M[A, A]$.

4. TRANSLATE COVERS AND EQUALITY

For a nonempty subset $A \subseteq G$, define

$$\tau_L(A) = \min\{|T| : T \subseteq G, TA = G\}, \quad \tau_R(A) = \min\{|T| : T \subseteq G, AT = G\}.$$

Theorem 4.1 (Two-sided translate covers). *Under the hypotheses of Theorem 1.1,*

$$\tau_L(A) \leq 2^n, \quad \tau_R(A) \leq 2^n.$$

More precisely, in the normalized construction both numbers are at most rank M .

Proof. Let $x_1, \dots, x_r$ index actual basis rows of M , where $r = \text{rank } M$. Their supports cover all columns by Lemma 2.2; by Equation (3.1), these supports are $x_1A, \dots, x_rA$. Hence $G = \bigcup_{j=1}^r x_jA$.

For fixed y , the support of column y is

$$\{x : x^{-1}y \in A\} = yA^{-1}.$$

Actual basis columns therefore give $G = \bigcup_j y_jA^{-1}$; taking inverses gives $G = \bigcup_j Ay_j^{-1}$. Since $r \leq 2^n$, the normalized bounds follow. If $A' = g_0^{-1}A$ is the normalized complement, then $TA' = G$ implies $(Tg_0^{-1})A = G$, while $A'T' = G$ implies $AT' = G$. $\square$

As a counting certificate, choose one representation $g = t(g)a(g)$ from a left cover $TA = G$. Then $g \mapsto (t(g), a(g))$ injects G into $T \times A$.

Theorem 4.2 (Equality rigidity). *Under the hypotheses of Theorem 1.1, if $|A| = |G|/2^n$, then A is a subgroup coset of index 2^n . After normalization by a point of A , the complement itself is a subgroup.*

Proof. Work in the normalized setting $e \in A$ and put $m = |A|$, $N = |G|$. Equality forces

$$\text{rank } M = 2^n = N/m.$$

Any basis of actual rows has N/m supports, each of size m , covering G . Their total size is N , so they are pairwise disjoint.

Suppose rows indexed by x, y have overlapping supports. If they were independent, then, because the actual rows span the row space, they could be extended to a basis of actual rows, contradicting the preceding disjointness. The two rows are therefore dependent; neither is zero, so they are proportional and have equal supports. Consequently

$$(4.1) \quad xA \cap yA \neq \emptyset \implies xA = yA.$$

Let $H = \{g \in G : gA = A\}$. If $a \in A$, then $a \in A \cap aA$ because $e \in A$, and Equation (4.1) gives $aA = A$; hence $A \subseteq H$. Conversely, $h \in H$ implies $h = he \in hA = A$, so $H \subseteq A$. Thus $A = H$ is a subgroup. Undoing normalization makes the original complement a left coset of H , and $[G : H] = |G|/|A| = 2^n$. $\square$

Remark 4.3. Theorem 4.2 classifies the uncovered set, not the covering family. Classification of all families of n left or right subgroup cosets whose union is the complement of a subgroup coset of index 2^n remains open.

5. COMPATIBLE BLOCKS

Definition 5.1. A compatible block of size $b \geq 1$ is either b left cosets $a_1H, \dots, a_bH$ of one subgroup H , or b right cosets $Ha_1, \dots, Ha_b$ of one subgroup H . A compatible block partition is a partition of the given coset family into such blocks.

Lemma 5.2 (Rank- $(b+1)$ block detector). *Every compatible block of size b admits a rational matrix of rank at most $b+1$ whose zeros are exactly the pairs (x, y) for which $x^{-1}y$ lies in one of the cosets in that block.*

Proof. For a left block $a_1H, \dots, a_bH$, choose an injection $\lambda : G/H \rightarrow \mathbb{Q}$ and define

$$L_H(x, y) = \prod_{j=1}^b (\lambda(yH) - \lambda(xa_jH)).$$

The j th factor vanishes exactly when $x^{-1}y \in a_jH$. Expanding as a polynomial of degree b in the single column variable $t = \lambda(yH)$ gives $L_H(x, y) = \sum_{r=0}^b u_r(x)t^r$, a sum of $b+1$ outer products. Hence $\text{rank } L_H \leq b+1$.

For a right block $Ha_1, \dots, Ha_b$, choose an injection $\mu : G/H \rightarrow \mathbb{Q}$ and put

$$R_H(x, y) = \prod_{j=1}^b (\mu(xH) - \mu(ya_j^{-1}H)).$$

Here

$$x^{-1}y \in Ha_j \iff x^{-1}ya_j^{-1} \in H \iff ya_j^{-1}H = xH.$$

Thus the zero set is again exact, and expansion in the single row variable $u = \mu(xH)$ gives rank at most $b+1$. $\square$

Theorem 5.3 (Compatible-block refinement). *Suppose $t \geq 1$ and the forbidden cosets have a compatible block partition $\mathcal{B}_1, \dots, \mathcal{B}_t$, where $b_j = |\mathcal{B}_j|$, and set*

$$R = \prod_{j=1}^t (b_j + 1).$$

If $A = G \setminus \bigcup_i C_i$ is nonempty, then

$$|A| \geq \frac{|G|}{R}, \quad \tau_L(A) \leq R, \quad \tau_R(A) \leq R.$$

If $|A| = |G|/R$, then A is a subgroup coset of index R .

Proof. Take one matrix from Lemma 5.2 for each block and form their Hadamard product B . Its exact support is $B(x, y) \neq 0 \iff x^{-1}y \in A$, and $\text{rank } B \leq R$ by Lemma 2.1. Choose $z \in A$. The entries $B(x, xz)$ form a nonzero permutation transversal, and each row has exactly $|A|$ nonzeros. Hence Lemma 2.2 gives $|G| \leq |A| \text{rank } B \leq |A|R$. Its row- and column-support conclusions give the two translate-cover bounds as in Theorem 4.1.

If equality holds, then $\text{rank } B = R = |G|/|A|$. The proof of Theorem 4.2, using the supports xA , applies verbatim after translating one element of A to the identity. Thus A is a subgroup coset of index R . $\square$

Singleton blocks recover $R = 2^n$, while a block of size $b \geq 2$ replaces 2^b by the smaller factor $b+1$. Repeated identical cosets may first be deleted. One may minimize R over all compatible block partitions of the given description.

6. SHARPNESS AND EQUALITY EXAMPLES

The grouped denominator is attained by the standard coordinate-fiber construction, familiar from finite-grid and abelian coset-cover arguments [AF93; Sze07; LS08; ST22].

Proposition 6.1 (Coordinate-fiber sharpness). *For every tuple of positive integers $(b_1, \dots, b_t)$ and every finite group K , the denominator $R = \prod_j (b_j + 1)$ in Theorem 5.3 is attained. In particular, the constant 2^{-n} in Theorem 1.1 is sharp.*

Proof. Take

$$G = K \times \prod_{j=1}^t C_{b_j+1}.$$

For coordinate j , use the b_j nonzero fibers, each a coset of the coordinate-zero subgroup. Their union misses exactly $K \times \{(0, \dots, 0)\}$, which has relative size $1/R$. Setting every $b_j = 1$ gives $G = K \times (C_2)^n$ and the sharp constant 2^{-n} . $\square$

The next transparent equality family is the index-two specialization of Sun's stronger exact-cover construction [Sun01, Example 1.2, p. 417]; see also [Sun06]. It is recorded only to contrast disjoint and overlapping equality mechanisms.

Proposition 6.2 (Sun's subgroup-chain example, index-two case). *Suppose*

$$G = H_0 > H_1 > \cdots > H_n, \quad [H_{j-1} : H_j] = 2,$$

and choose $a_j \in H_{j-1} \setminus H_j$. Then

$$G \setminus H_n = \bigsqcup_{j=1}^n a_j H_j.$$

Thus these n cosets attain equality in [Theorem 1.1](#).

Proof. For each j , the two left cosets of H_j in H_{j-1} are H_j and $a_j H_j$, so $H_{j-1} \setminus H_j = a_j H_j$. The successive differences are disjoint and telescope to $G \setminus H_n$; multiplying the indices gives $[G : H_n] = 2^n$. $\square$

The coordinate construction has substantial overlap among the forbidden cosets, whereas the subgroup-chain layers are disjoint. This already rules out any equality-family classification based only on one of those two patterns.

APPENDIX A. COMPACT CONSEQUENCES

A.1. An abstract rank-support formulation. For convenience we combine the two standard rank facts above in one form.

Theorem A.1 (Rank-support incidence principle). *Let X and Y be finite sets of the same positive size N , let $s \geq 1$, and for $1 \leq i \leq s$ let B_i be an $X \times Y$ matrix over a field $\mathbb{F}$ with $\text{rank } B_i \leq r_i$. Set $B = B_1 \circ \cdots \circ B_s$. Suppose every row of B has at most m nonzeros and there is a bijection $\pi : X \rightarrow Y$ with $B(x, \pi(x)) \neq 0$ for all x . Then*

$$N \leq m \text{rank } B \leq m \prod_{i=1}^s r_i.$$

At most $\prod_i r_i$ actual row supports cover Y , and at most $\prod_i r_i$ actual column supports cover X .

Proof. Apply [Lemmas 2.1](#) and [2.2](#). The transversal forces $m \geq 1$ and $\text{rank } B > 0$, so no empty-product convention is involved. $\square$

A.2. Cayley exact-support rank. Exact zero–nonzero pattern rank was studied before the present setting [HS93]. De Wolf's nondeterministic rank minimizes complex rank over matrices nonzero exactly on the 1-inputs of a Boolean function [Wol00]; the following is its Cayley-support specialization, with an arbitrary ground field. Unlike graph or Haemers minrank conventions, every position allowed by the support relation is required to be nonzero; this distinction is discussed in [DY26].

Definition A.2. For a nonempty subset A of a finite group G and a field $\mathbb{F}$, define

$$\text{mr}_{\mathbb{F}}(A) = \min\{\text{rank } B : B(x, y) \neq 0 \iff x^{-1}y \in A\},$$

where rows and columns are indexed by G .

Row and column permutations show that $\text{mr}_{\mathbb{F}}(sA) = \text{mr}_{\mathbb{F}}(A) = \text{mr}_{\mathbb{F}}(At)$ for all $s, t \in G$.

Theorem A.3 (Exact-support bounds and equality). *Let A be a nonempty subset of a finite group G .*

- (a) *For every field $\mathbb{F}$, $\text{mr}_{\mathbb{F}}(A) \geq |G|/|A|$.*
- (b) *Equality holds if and only if A is a subgroup coset.*
- (c) *If $G \setminus A$ is a union of n left or right subgroup cosets, then $\text{mr}_{\mathbb{Q}}(A) \leq 2^n$. A compatible block partition with denominator R gives $\text{mr}_{\mathbb{Q}}(A) \leq R$.*

Proof. Choose an exact-support matrix B of rank $\text{mr}_{\mathbb{F}}(A)$. For any $a \in A$, it has the nonzero transversal (x, xa) and exactly $|A|$ nonzeros in every row, proving (a) by Lemma 2.2. If equality holds, every actual row basis of B has $|G|/|A|$ supports of size $|A|$ covering G , so those supports are disjoint. The disjoint-or-equal and stabilizer argument in Theorem 4.2 then shows, after translating one element of A to the identity, that A is a subgroup coset. Conversely, if $A = aH$, the 0–1 exact-support matrix has as its distinct rows the indicators of the left cosets of H ; those disjoint nonzero rows have rank $[G : H] = |G|/|A|$ over every field. Part (c) is supplied by the detectors in Theorems 1.1 and 5.3. $\square$

Part (a) is the elementary sparse-transversal bound. No historical novelty is claimed for the equality characterization in (b); its priority in the full arbitrary-completion setting remains unresolved. The upper bounds in (c) are detector-derived applications not located in the targeted search. Related support–dimension inequalities and subgroup-coset equality phenomena for orbit-generated or convolutional matrices appear in [GGI05; BWZ22; CFL25], but those results do not automatically apply to an arbitrary exact-support completion.

For any nonempty $A \subseteq G$, let $\kappa_G(A)$ be the minimum number of left or right subgroup cosets whose union is $G \setminus A$, with the empty-family convention $\kappa_G(G) = 0$. If $A = G$, then $\text{mr}_{\mathbb{Q}}(G) = 1$ and $\kappa_G(G) = 0$. Otherwise Theorem A.3(c) gives the immediate obstruction

$$\kappa_G(A) \geq \lceil \log_2 \text{mr}_{\mathbb{Q}}(A) \rceil.$$

A.3. Prescribed images and derangements. Let a finite group G act on a finite set Ω , and fix pairs $(p_i, q_i) \in \Omega^2$ for $1 \leq i \leq n$. For $n = 0$ the avoiding set is G and all conclusions below are immediate, so assume $n \geq 1$. Each nonempty fiber $\{g : gp_i = q_i\}$ is a left coset of the stabilizer G_{p_i} . Consequently the set

$$D = \{g \in G : gp_i \neq q_i \text{ for every } i\}$$

is empty or satisfies

$$|D| \geq |G|/2^n, \quad \tau_L(D), \tau_R(D) \leq 2^n.$$

If the nonempty constraints are grouped by source point and source p_j has b_j distinct reachable forbidden images, then 2^n may be replaced by $\prod_j (b_j + 1)$. These are direct point-stabilizer applications, not independent permutation-group methods.

For derangements, put $\text{Der}(G, \Omega) = \{g : gp \neq p \text{ for every } p \in \Omega\}$. If $\Omega = \emptyset$, then $\text{Der}(G, \Omega) = G$ directly and the bound and one-translate covers hold with $2^{|\Omega|} = 1$. If $\Omega \neq \emptyset$, the preceding result gives either an empty derangement set or

$$|\text{Der}(G, \Omega)| \geq |G|/2^{|\Omega|}, \quad \tau_L(\text{Der}), \tau_R(\text{Der}) \leq 2^{|\Omega|}.$$

For a nontrivial transitive action of degree d and permutation rank r , Cameron and Cohen proved the substantially stronger estimate

$$\frac{|\text{Der}(G, \Omega)|}{|G|} \geq \frac{r-1}{d} \geq \frac{1}{d},$$

which is strictly stronger than 2^{-d} for $d \geq 2$ [CC92, p. 136]; see also later refinements in [BS24]. The present statement is recorded only for its uniform prescribed-image formulation.

A.4. Finite-index and profinite transfers. The common-normal-core mechanism for finite unions of mixed left and right finite-index cosets is standard; compare the classical coset-cover framework of [Neu54] and the explicit common-core treatment in Button [But15, Propositions 2.4–2.6].

Proposition A.4. *Let G be any group, let $n \geq 1$, and let $C_1, \dots, C_n$ be left or right cosets of finite-index subgroups. Suppose $A = G \setminus \bigcup_i C_i$ is nonempty. There is a finite-index normal subgroup $N \triangleleft G$ such that A is a union of N -cosets and*

$$\frac{|A/N|}{|G/N|} \geq 2^{-n},$$

where A/N denotes the image of A in G/N . Moreover $G = T_L A = A T_R$ for finite sets T_L, T_R with $|T_L|, |T_R| \leq 2^n$. A compatible block denominator R replaces 2^n when available.

Proof. Let H_i be the subgroup underlying C_i and set $N = \bigcap_i \text{core}_G(H_i)$, where $\text{core}_G(H_i) = \bigcap_{g \in G} g H_i g^{-1}$. Each core, and hence N , is finite-index normal. Since $N \leq H_i$, every C_i and A are full inverse images of sets in the finite quotient G/N , with the original left or right orientation retained. Apply [Theorems 1.1, 4.1 and 5.3](#) in the quotient and lift representatives of the finite left and right translate covers. $\square$

If G is profinite and the C_i are cosets of open subgroups, the same N is open normal. The complement is clopen and descends to the finite quotient; normalized Haar measure is the quotient proportion, a standard profinite fact [[Seg07](#), pp. 24–25]. Therefore a nonempty complement satisfies $\mu(A) \geq 2^{-n}$ and has left and right translate-cover number at most 2^n , with the compatible-block denominator R available as before. These infinite and profinite statements are routine transfers of the finite constants.

APPENDIX B. COMPUTATION AND OPEN QUESTIONS

The proof is independent of computation. The accompanying exact verifier uses explicit multiplication tables and rational Gaussian elimination. Its fresh replay exhausts all families of at most three distinct proper coset sets in C_6 , S_3 , D_8 , Q_8 , and $(C_2)^3$; checks every equality complement in that census; verifies a mixed left/right D_8 detector and a nonnormal right block; and checks grouped sharpness for profiles $(1, 1, 1)$ and $(2, 3)$. The replay examines 29,505 families, including 28,773 proper unions and 3,070 equality cases, and ends with all assertions passing. These finite checks are regression tests, not a proof.

Question B.1 (Equality families). Classify all equality-achieving families of n left or right subgroup cosets. Equivalently, once [Theorem 4.2](#) identifies the complement as a subgroup coset of index 2^n , classify all n -coset representations of its complement.

Question B.2 (Stability). If $|A|$ is only slightly larger than $|G|/2^n$, under what hypotheses must A be close to a subgroup coset? Can a quantitative conclusion be extracted from near-equality in the support-cover rank argument?

Question B.3 (Further compression). Can compatible blocks, subgroup chains, repeated quotient coordinates, and other algebraic dependencies be unified by a canonical exact-support rank parameter sharper than $\prod_j (b_j + 1)$?

VERIFICATION AND PROVENANCE

DannyExperiments initiated and directed the investigation, selected the problem and validation requirements, curated and validated the artifacts, and is the human curator and conventional manuscript author of this project. OpenAI GPT-5.6 Pro generated the ordinary-language proof, the strengthened results, and the manuscript draft. OpenAI Codex produced one independent Lean 4 formalization, coordinated the audit and provenance workflow, and prepared the repository and reproducibility infrastructure. Aristotle produced a second independent Lean 4 formalization.

The audited Aristotle-derived Lean 4.30 port and the independent Codex Lean 4.30 certificate verify only the headline mixed-left/right finite-group complement inequality; none of the strengthenings is claimed to be formally verified. No AI system is listed as an author. The preserved mathematical and literature reports are AI-generated audits, not human peer review. Human specialist review has not been obtained. The documented literature search supports only the qualified classification “apparently new, moderate confidence”; absolute historical priority is not claimed.

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